Loops and Stopping

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Loops and Stopping before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Trace Tables

Without opening the old lesson, explain one key idea from **Trace Table**s. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Decompose a Goal

Without opening the old lesson, explain one key idea from Decompose a Goal. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Loops and Stopping trace and record the first step you can explain without help.

Use W2: Trace before running for **Loop**s and **trace table**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Loops and Stopping explanation against the supplied facts before

discussing it.

Use the worked case for **Loop**s and **trace table**s. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Changed independent case

Start count at 1. While count < 8, add 3. Trace the condition and values. What happens if the condition becomes count != 8?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Loops and Stopping target to a trace and a justified conclusion.

> Trace count=0; while count<4: count=count+2. List values tested by the condition and final count. Would replacing <4 with !=5 terminate?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Loops and Stopping case and show the deciding evidence.

Start n=0; while n<3, add 1 and record n.

List records and stopping condition.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Start at count 0. While count is less than 3, add 1 and display count. List the displays and the stopping value.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 5 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Loops and Stopping adds to Transparent Book Recommender and keep the evidence.

Loop through every candidate book, **update** its score, then stop at the end of the list.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

> Trace count=0; while count<4: count=count+2. List values tested by the condition and final count. Would replacing <4 with !=5 terminate?

> **My independently reasoned answer**

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Explain the range

> Explain the range. Apply the supplied Loops and Stopping facts and keep your reasoning.

List the values supplied by range(2, 5) and explain why 5 is absent. How many times does its body run?

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Trace before running

> Trace before running. Apply the supplied Loops and Stopping facts and keep your reasoning.

Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Repair the reset

> Repair the reset. Apply the supplied Loops and Stopping facts and keep your reasoning.

For the faulty script with range(1, 4), record total after each addition. Explain the wrong final **output** and remove only the faulty reset.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Build a repeated addition independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Build a repeated addition independently. Apply the supplied Loops and Stopping facts and keep your reasoning.

A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the **loop**.

Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Check a contract limitation

> Check a contract limitation. Apply the supplied Loops and Stopping facts and keep your reasoning.

In your counter script, range(-2) performs no **iteration**s. Does returning the initial 5 make negative rounds valid? Explain what the script checks and what it assumes.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Loops and Stopping

> A fictional counter starts at 12 and decreases while above a boundary. Explain the labelled step before checking the changed condition.

A fictional counter starts at 12 and decreases while above a boundary.

Python: count=12; while count>3: count=count-4; print after the **loop**.

The condition is checked before each subtraction.

AI ACADEMY / AI-2/5

Loops and Stopping

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Initial check

12>3: true; subtract 4 to get 8.



2. Next check

8>3: true; subtract 4 to get 4.



3. Final update

4>3: true; subtract 4 to get 0.



4. Stop

0>3 is false; report 0 after three updates.

CHANGE ONE THING

Replace count>3 with count!=3.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why is report 0 correct for the original **loop**, and why is equality-only stopping risky here?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Use progress and stopping arguments for any repeated recommender **process**; test **input**s that require zero, one and several **update**s.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use progress and stopping arguments for any repeated recommender **process**; test **input**s that require zero, one and several **update**s.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional counter starts at 12 and decreases while above a boundary.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? 0>3 is false; report 0 after three **update**s.

> 1. Next check

2. Final **update**

3. Stop

4. Initial check

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Stop 0>3 is false; report 0 after three **update**s. The condition does not promise the counter ends exactly at 3. Subtractions can cross the boundary.

The count strictly decreases, so the below-or-equal stopping region is eventually reached from this start. A condition requiring exact equality can be unsafe when repeated **update**s skip that value.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace count>3 with count!=3. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. From 12, the **loop** never reaches 3 and does not terminate in this toy model.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: From 12, the **loop** never reaches 3 and does not terminate in this toy model. 12,8,4,0,−4,… are multiples of 4; none equals 3.

Repair a missing building block

Review **Trace Table**s (ai-2/4). Goal: Trace variable values before and after each **loop iteration**. Short practice: Move print inside the original **loop**. What sequence is displayed?.

Return to this week and explain what you changed.

Review Decompose a Goal (ai-2/2). Goal: Distinguish a function definition from a function call. Short practice: What happens visibly if the caller stores needed but omits print(needed)?.

Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

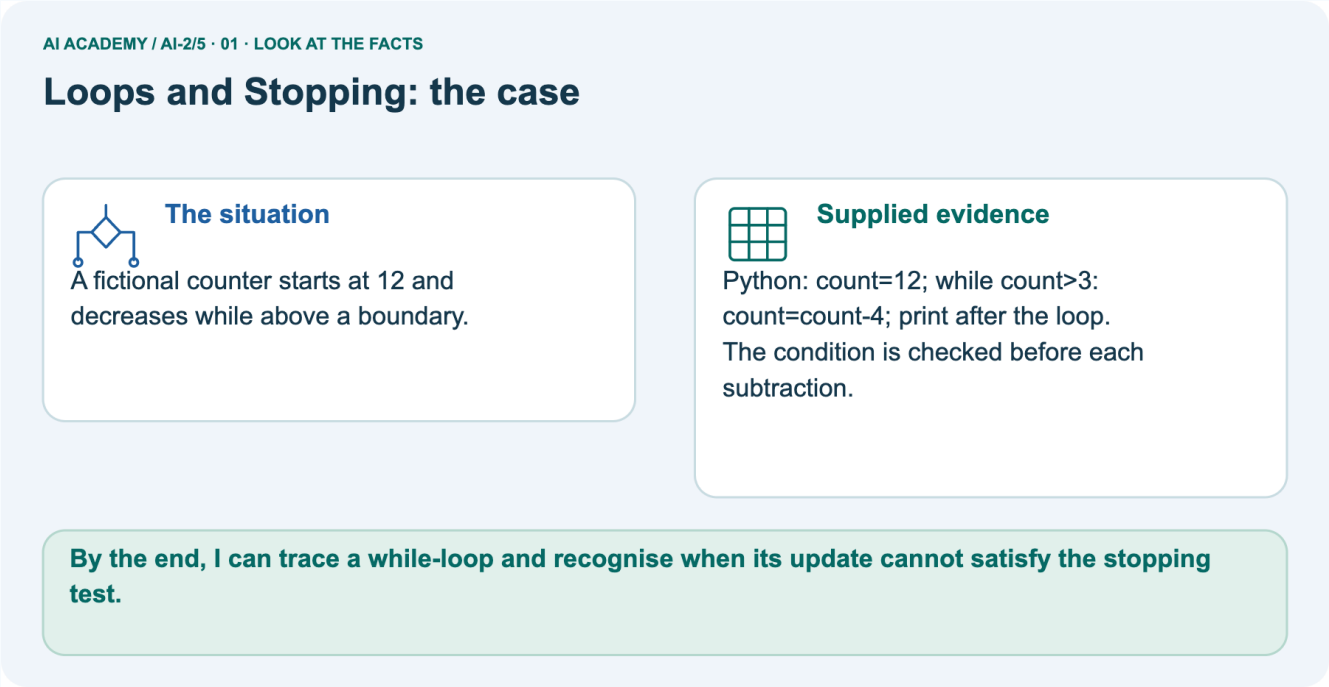
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Loops and Stopping picture from its labels, then record a relevant limit.

01. Loops and Stopping: the case



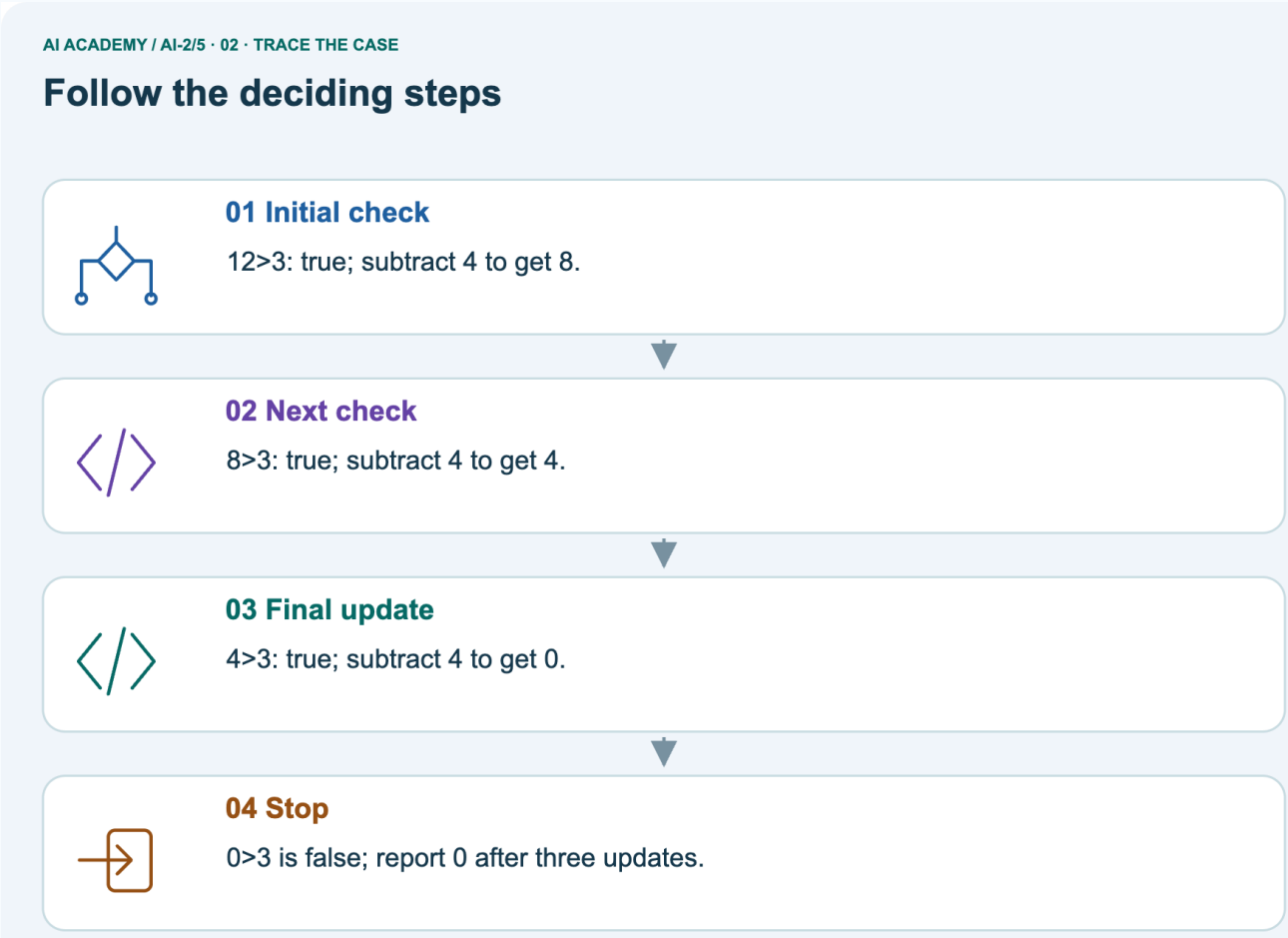
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional counter starts at 12 and decreases while above a boundary. Supplied facts: Python: count=12; while count>3: count=count-4; print after the **loop**. The condition is checked before each subtraction.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Initial check: 12>3: true; subtract 4 to get 8. → Next check: 8>3: true; subtract 4 to get 4. → Final **update**: 4>3: true; subtract 4 to get 0. → Stop: 0>3 is false; report 0 after three **update**s.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. State after each addition

AI ACADEMY / AI-2/5 · 03 · INSPECT THE LESSON MODEL

State after each addition

turn	total before	total after
1	0	1
2	1	3
3	3	6

Initialize total to 0 before the loop. The unindented print runs once afterward.

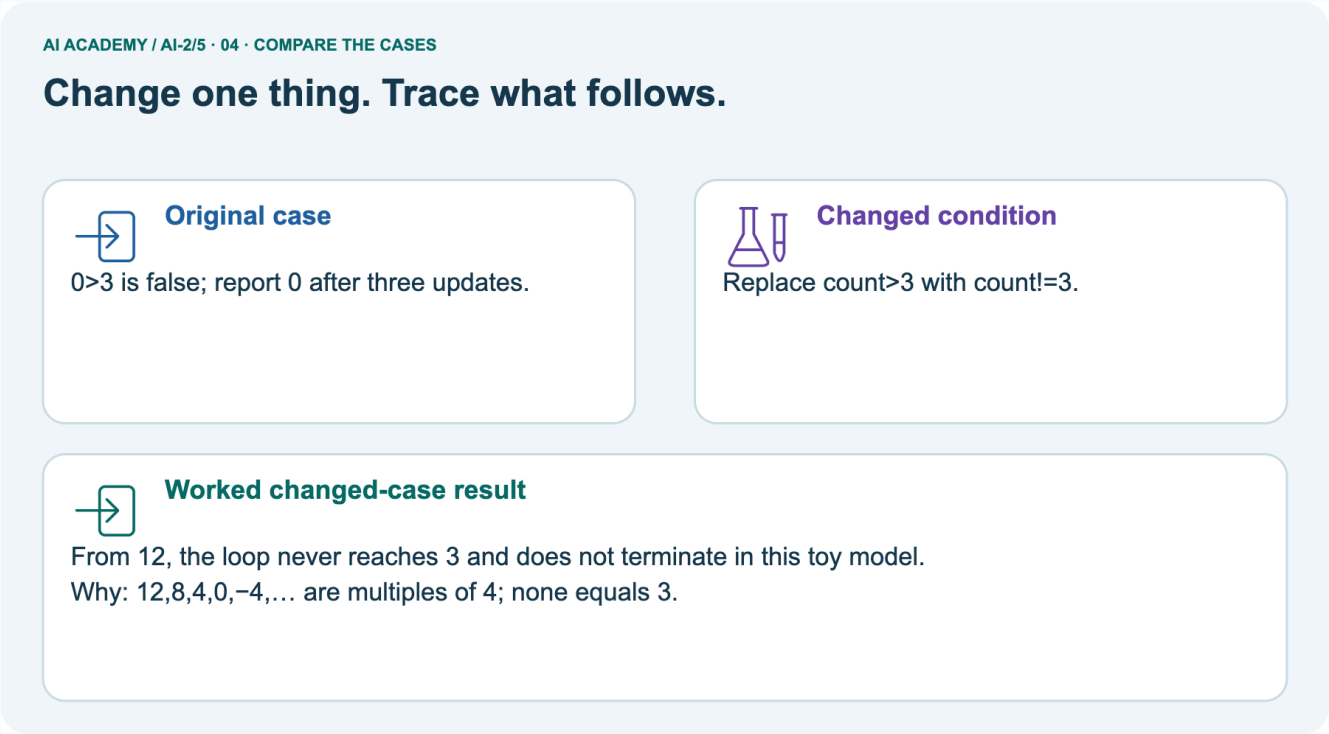
Follow the labelled picture with the matching explanation. Open the vector diagram

State after each addition. 1 | 0 | 1. 2 | 1 | 3. 3 | 3 | 6

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

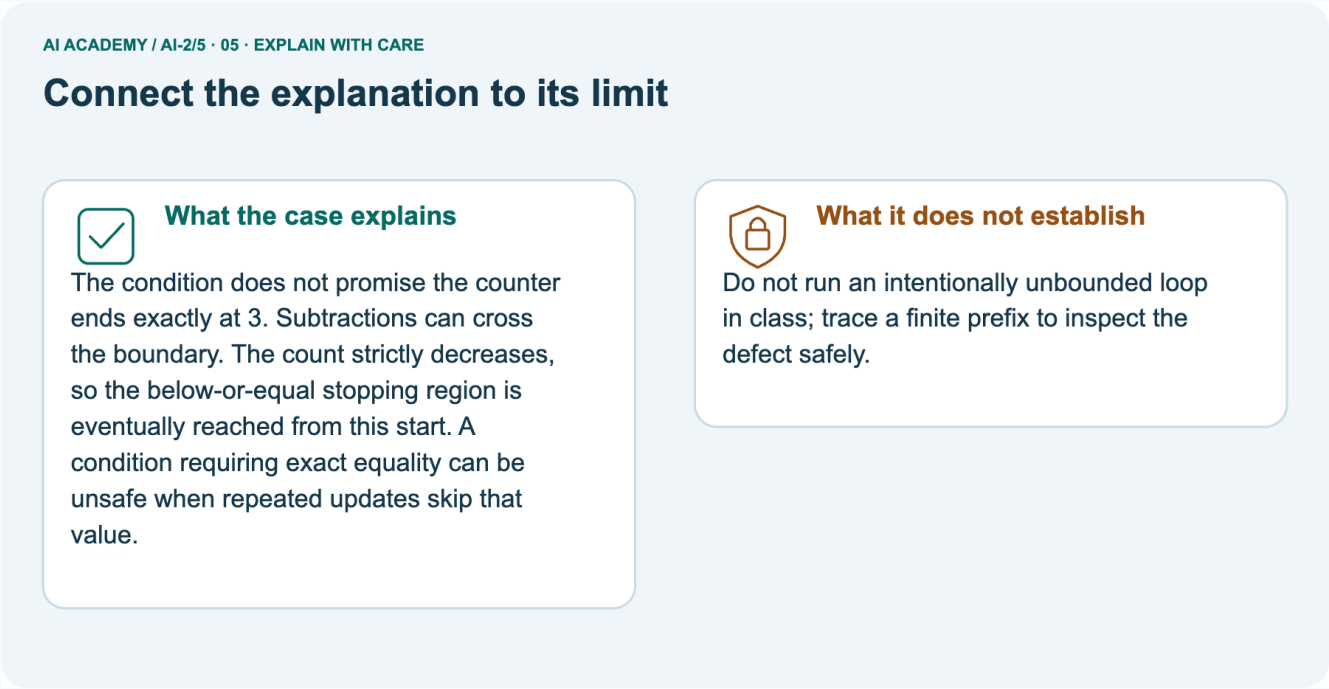
Original case: 0>3 is false; report 0 after three **update**s. Changed condition: Replace count>3 with count!=3. Expected result: From 12, the **loop** never reaches 3 and does not terminate in this toy model.

Reason: 12,8,4,0,−4,… are multiples of 4; none equals 3.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

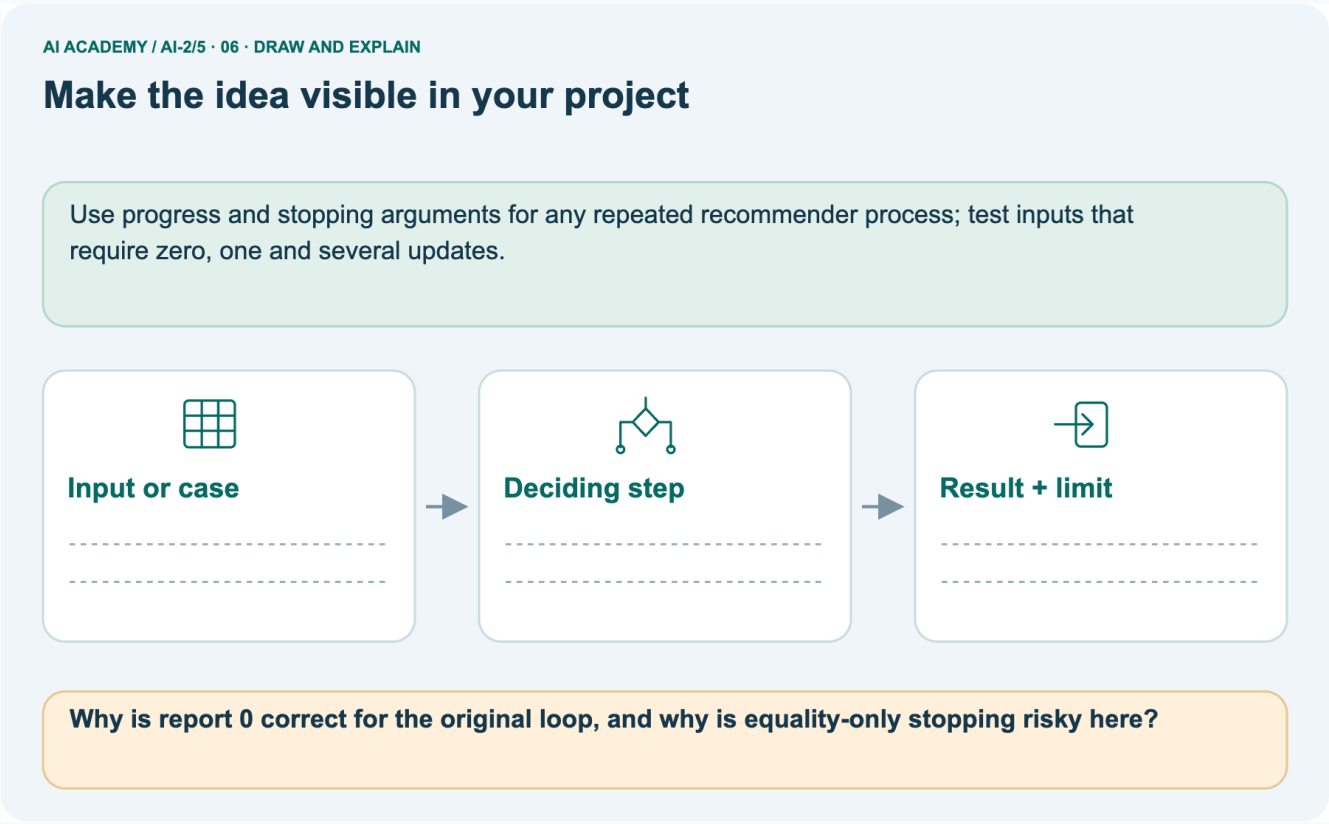
The condition does not promise the counter ends exactly at 3. Subtractions can cross the boundary. The count strictly decreases, so the below-or-equal stopping region is eventually reached from this start.

A condition requiring exact equality can be unsafe when repeated **update**s skip that value. Limit: Do not run an intentionally unbounded **loop** in class; trace a finite prefix to inspect the defect safely.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use progress and stopping arguments for any repeated recommender **process**; test **input**s that require zero, one and several **update**s. Draw your **input**, deciding step, result and limit.

Supported question: Why is report 0 correct for the original **loop**, and why is equality-only stopping risky here?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.